# Standard Operating Procedure for Resolving Optical Transport Failure on Huawei Access Nodes (JAY-ACC-39) at Jayanagar Metro Site

**Node:** JAY-ACC-39 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_3328

## Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Nodes (Role: Access, Node ID: JAY-ACC-39, Site Location: Jayanagar Metro Site) is primarily attributed to the degradation of the SFP/XFP transceivers or physical damage to the fiber optic cables. This failure domain is specific to this role and vendor equipment due to the underlying protocol mechanics. The Huawei Access Node utilizes the Optical Transport Network (OTN) protocol for wavelength division multiplexing (WDM) and optical transmission. In the event of SFP/XFP degradation or fiber cuts, the OTN protocol may not be able to detect the failure immediately, leading to a warning alarm (Alarm Code: ALARM_3328, Severity: Warning). Therefore, it is essential to promptly diagnose and resolve the issue to prevent any potential network downtime or service disruption.

## Pre-requisite Checks

- check: Power up the Node (JAY-ACC-39) and verify that all redundant links are functioning properly before proceeding with diagnostic and resolution steps.

- check: Ensure remote hands are dispatched to the Jayanagar Metro Site for assistance, if necessary.

## Potential Root Causes

| Cause | Probability |
|---|---|
| Degradation of SFP/XFP transceivers | High |
| Physical damage to fiber optic cables | Medium |
| OTN protocol misconfiguration | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display optical-transceiver info, Expected Output: Transceiver information: SFP/XFP status, wavelength, and optical power., Description: This command displays the current status of the SFP/XFP transceivers, including any degradation or optical power loss. This information helps identify potential issues with the transceivers or fiber optic cables.

Command: display optical-fiber info, Expected Output: Fiber information: fiber type, length, and status., Description: This command displays information about the fiber optic cables, including their type, length, and status. This information helps identify any physical damage or issues with the fiber optic cables.
```

```
Command: display otu-mpls info, Expected Output: OTN/MPLS information: label distribution, QoS, and forwarding information., Description: This command displays OTN/MPLS information, including label distribution, QoS, and forwarding information. This information helps identify any OTN protocol misconfiguration or issues with the OTN/MPLS network.
```

## Resolution Steps

Step 1: Step: Replace degraded SFP/XFP transceivers, Command: replace optical-transceiver {transceiver-name}, Description: Replace the degraded SFP/XFP transceivers with new ones. Ensure that the replacement transceivers meet the required specifications and are properly configured.

Step 2: Step: Repair or replace damaged fiber optic cables, Command: repair optical-fiber {fiber-name}, Description: Repair or replace the damaged fiber optic cables. Ensure that the replacement cables meet the required specifications and are properly installed.

Step 3: Step: Verify OTN protocol configuration, Command: display otu-mpls config, Description: Verify the OTN protocol configuration, including label distribution, QoS, and forwarding information. Ensure that the configuration is correct and consistent across the OTN/MPLS network.

## Post-Check Verification

```
Command: display optical-transceiver info, Expected Output: Transceiver information: SFP/XFP status, wavelength, and optical power., Description: Verify that the SFP/XFP transceivers are functioning properly and that the optical power is within the required range.

Command: display optical-fiber info, Expected Output: Fiber information: fiber type, length, and status., Description: Verify that the fiber optic cables are functioning properly and that there is no physical damage or issues with the cables.

Command: display otu-mpls info, Expected Output: OTN/MPLS information: label distribution, QoS, and forwarding information., Description: Verify that the OTN/MPLS network is functioning properly and that there are no issues with label distribution, QoS, or forwarding information.
```

## Roll-back Procedures

- Step: Restore original SFP/XFP transceivers, Command: restore optical-transceiver {transceiver-name}, Description: Restore the original SFP/XFP transceivers if the replacement transceivers do not resolve the issue.

- Step: Restore original fiber optic cables, Command: restore optical-fiber {fiber-name}, Description: Restore the original fiber optic cables if the replacement cables do not resolve the issue.

- Step: Restore original OTN protocol configuration, Command: restore otu-mpls config, Description: Restore the original OTN protocol configuration if the replacement configuration does not resolve the issue.